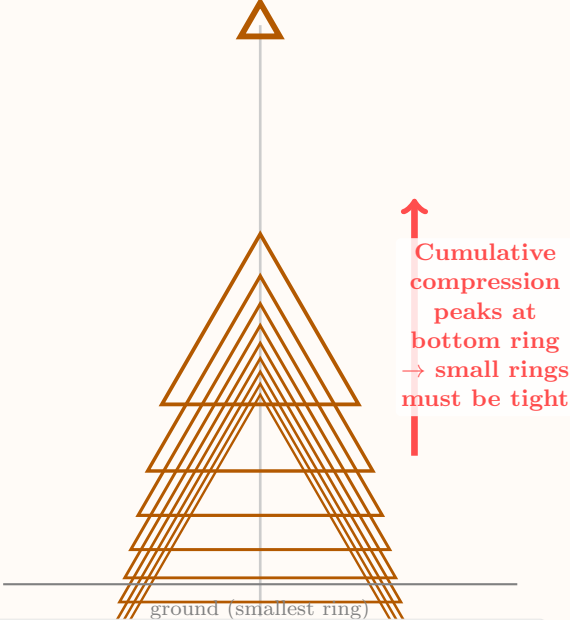


Why the Density Profile Inverted: From Bottom-Heavy to Top-Biased

At n=3, per-beam compression is high — rings must cluster at the bottom. At n=12, load is distributed — the sign flips.

n = 3, $\beta = +0.76$
Heavily bottom-biased — rings cluster at base



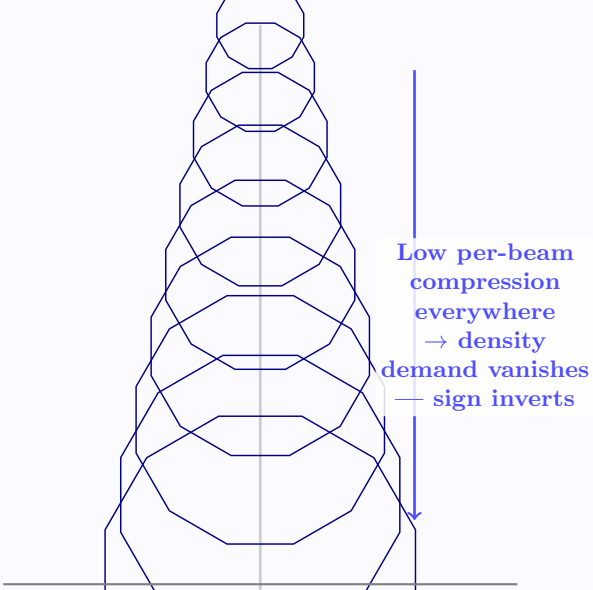
Per-beam compression: $N/3$ — high
Bottom ring radius = 0.35, top ring = 2.75
Why: Small rings at bottom resist high per-beam load

Cumulative
compression
peaks at
bottom ring
→ small rings
must be tight

n=3 → n=12

density sign flips

n = 12, $\beta = -0.13$
Mild top-bias — sign has inverted



Per-beam compression: $N/12$ — $4\times$ lower
Bottom ring radius = 0.7, top ring = 2.5
Why $\beta < 0$: Compression so low that beam taper (thin at hub) now dominates over load — rings drift slightly toward the top where beams are thinnest.

Low per-beam
compression
everywhere
→ density
demand vanishes
— sign inverts